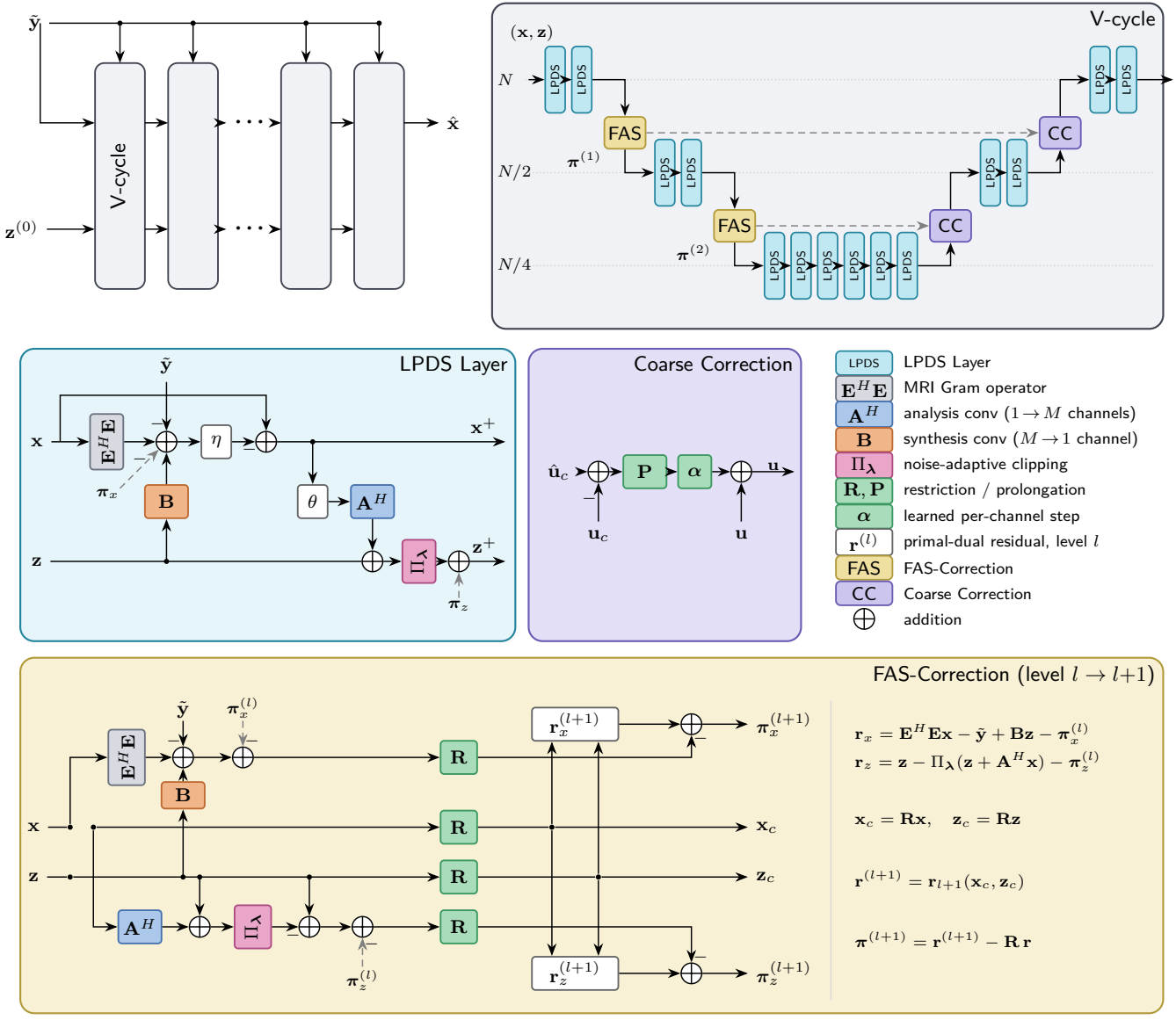

$\tilde{\mathbf{y}}$
V-cycle
$\hat{\mathbf{x}}$
$\mathbf{z}^{(0)}$
V-cycle
$(\mathbf{x}, \mathbf{z})$
N
$N/2$
$N/4$
LPDS
FAS
CC
$\boldsymbol{\pi}^{(1)}$
$\boldsymbol{\pi}^{(2)}$
LPDS Layer
$\mathbf{x}$
$\mathbf{z}$
$\mathbf{E}^H\mathbf{E}$
η
θ
$\mathbf{A}^H$
$\mathbf{B}$
$\Pi_{\boldsymbol{\lambda}}$
$\boldsymbol{\pi}_x$
$\boldsymbol{\pi}_z$
$\mathbf{x}^+$
$\mathbf{z}^+$
Coarse Correction
$\hat{\mathbf{u}}_c$
$\mathbf{u}_c$
$\mathbf{P}$
$\boldsymbol{\alpha}$
$\mathbf{u}$
LPDS Layer
$\mathbf{E}^H\mathbf{E}$ MRI Gram operator
$\mathbf{A}^H$ analysis conv ($1 \to M$ channels)
$\mathbf{B}$ synthesis conv ($M \to 1$ channel)
$\Pi_{\boldsymbol{\lambda}}$ noise-adaptive clipping
$\mathbf{R}, \mathbf{P}$ restriction / prolongation
$\boldsymbol{\alpha}$ learned per-channel step
$\mathbf{r}^{(l)}$ primal-dual residual, level l
FAS FAS-Correction
CC Coarse Correction
addition
FAS-Correction (level $l \to l{+}1$)
$\mathbf{r}_x^{(l+1)}$
$\mathbf{r}_z^{(l+1)}$
$\boldsymbol{\pi}_x^{(l+1)}$
$\boldsymbol{\pi}_z^{(l+1)}$
$\mathbf{x}_c$
$\mathbf{z}_c$
$\mathbf{r}_x = \mathbf{E}^H\mathbf{E}\mathbf{x} - \tilde{\mathbf{y}} + \mathbf{B}\mathbf{z} - \boldsymbol{\pi}_x^{(l)}$
$\mathbf{r}_z = \mathbf{z} - \Pi_{\boldsymbol{\lambda}}(\mathbf{z} + \mathbf{A}^H\mathbf{x}) - \boldsymbol{\pi}_z^{(l)}$
$\mathbf{x}_c = \mathbf{R}\mathbf{x}, \quad \mathbf{z}_c = \mathbf{R}\mathbf{z}$
$\mathbf{r}^{(l+1)} = \mathbf{r}_{l+1}(\mathbf{x}_c, \mathbf{z}_c)$
$\boldsymbol{\pi}^{(l+1)} = \mathbf{r}^{(l+1)} - \mathbf{R}\,\mathbf{r}$